Flood Risk Exposure of Slovenian Settlements: Workflow Report

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Goal

This project assesses which settlements in Slovenia are exposed to flood hazards and flood risks, and to what degree are these hazards/risks are actually a concern. It combines national settlement point dataset with official flood hazard zone polygons and a digital elevation model (DEM) to answer three questions: (1) which settlements lie within 1 km of a mapped flood area; (2) how is flood risk distributed across settlements when classified by hazard return period; and (3) whether settlement elevation correlates with flood risk classification.

Note: All data is directly pulled from sources within the python code (see the Data Acquisition section of the .ipynb file), but cached files are also enclosed within the submission in case that is more useful.

Input Datasets

Note: For retrieving information from Overpass API using predefined tags, https://wiki.openstreetmap.org/wiki/Map_features was used to specify what types of specific information was necessary to retrieve from within Slovenia's boundaries. The enclosed Jupyter Notebook includes code to pull this information directly from the API and other sources as defined below, but cached files are also included in the submission for convenience.

- **Settlements (points):** OpenStreetMap, via Overpass API
- **Flood hazard zones (polygons):** INSPIRE WFS, eprostor.gov.si, layer PS.FloodedArea
- **Digital Elevation Model (raster):** EU-DEM v1.1 25 m, Copernicus, with SRTM 30 m as fallback (sometimes the 25m worked but not reliably so the 30m was included as a failsafe for the project - the cached file should be the SRTM)

  Note: Tried downloading directly from Copernicus but the file was corrupting and just weird to deal with, so ended up adding the SRTM pathway. The cached file is enclosed in the submission is from the SRTM. I'd missed that I forgot to delete the direct link so apologies that this is still included - this is my explanation.
- **Water bodies, rivers and lakes (lines / polygons):** OpenStreetMap, via Overpass API
- **Slovenia national boundary (polygon):** GADM 4.1

Note: Direct data links are provided in the .ipynb file, as all data is directly pulled from these sources.

Processing Steps

Data acquisition and caching. All layers were downloaded once (boundary, settlements, flood zones, water bodies, DEM) and cached locally to avoid repeated API calls. Settlement points were queried from Overpass API using a bounding-box filter on place=city/town/village, then clipped to the Slovenia national boundary to remove cross-border nodes. The flood zone column names from the WFS service were normalized (return period, zone name) since they vary across the source. This made cross-comparisons for the final map visualization much simpler.

Reprojection. All vector layers were reprojected to EPSG:3794 (Slovenia 1996 / national grid), Slovenia's official projected coordinate system, ensuring consistent and accurate distance and area measurements for the buffer and zonal statistics steps.

Step 1. DEM clipping. The elevation raster was masked to the Slovenia boundary polygon using rasterio, producing a clipped DEM restricted to the area of interest and reducing processing volume for subsequent steps.

Step 2. Zonal statistics. Mean, minimum, maximum and standard deviation of elevation were computed for each flood hazard polygon using rasterstats.zonal_stats, characterizing the terrain profile of each flood zone.

Step 3. Spatial join. Settlement points were spatially joined against flood hazard polygons using an intersects predicate. Where a settlement matched multiple overlapping flood polygons, the match with the lowest (most severe) return period was kept, giving one risk record per settlement.

Step 4. Risk classification. Settlements were classified into Very High (return period <= 10 yr), High (<= 100 yr), Moderate (<= 500 yr) or No hazard, based on the return period of the matched flood polygon.

Step 5. Buffer / proximity analysis. Flood polygons were buffered by 1 km and dissolved into a single proximity zone. Settlements not already inside a flood polygon were tested against this buffer; matches were re-labelled as Proximity (<= 1 km), capturing settlements at indirect risk.

Visualization and summary. Results were rendered as a map (settlements colored by risk class, flood zones and water bodies overlaid) alongside summary charts, and a final printed summary table reporting counts per risk category and mean elevation for at-risk versus unaffected settlements.

Output

The notebook produces a clipped DEM raster, a flood hazard layer enriched with elevation statistics, a per-settlement risk classification table, a composite map/chart visualization, and a printed summary (total settlements analyzed, counts in each risk tier, and mean elevation comparison between at-risk and unaffected settlements). Together these directly answer the three research questions stated in the goal.